

Day 3 MERN Assignment

Assignment 1: Employee Portal Authentication

Business Scenario

Secure employee login.

Collection

users{name,email,password(hash),role}

Backend Requirements

- Register API
- Hash password with bcrypt
- Login API
- Generate JWT
- Authentication middleware
- Authorization middleware for admin
- Protected profile route

Frontend Requirements

- React Register
- React Login
- Store JWT
- Send Authorization header
- Logout

Expected Output

JWT login and protected dashboard.

Assignment 2: Student LMS

Business Scenario

LMS login with trainer/student roles.

Collection

students{name,email,password,role}

Backend Requirements

- Register/Login
- bcrypt
- JWT
- Trainer-only routes

Frontend Requirements

- Register/Login UI
- Role dashboard

Expected Output

Role-based access.

Assignment 3: Hospital Staff Portal

Business Scenario

Doctor and receptionist roles.

Collection

staff{name,email,password,role}

Backend Requirements

- Hash passwords
- JWT
- Doctor-only route

Frontend Requirements

- Login
- Protected pages

Expected Output

Unauthorized users blocked.

Assignment 4: Bank Customer Portal

Business Scenario

Secure banking access.

Collection

customers{name,email,password}

Backend Requirements

- Register/Login
- bcrypt
- JWT
- Protected account

Frontend Requirements

- Forms
- Account page

Expected Output

Account visible only after login.

Assignment 5: E-Commerce Admin Panel

Business Scenario

Admin/customer authorization.

Collection

users{name,email,password,role}

Backend Requirements

- Register/Login
- bcrypt
- JWT
- Admin product APIs

Frontend Requirements

- Admin & customer dashboards

Expected Output

Admin-only management.

Submission

- Use bcrypt for password hashing.
- Use JWT for login.
- Implement authentication and authorization middleware.
- React must send JWT in Authorization header.
- Upload to GitHub.